

Beyond Pareto Fronts: A Leximax Universal-Regret Core

for Robust Multicriteria Decision Support

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Abstract

Pareto optimality is a safety property, not a complete decision rule: an efficient set tells a decision maker (DM) which alternatives are not obviously bad, but offers no principled way to choose one. Practice fills this gap with *post hoc* devices—weighted sums, TOPSIS, compromise programming, knee selection, outranking thresholds—each of which reintroduces exactly the preference parameters the DM was unable to state. We propose the **Leximax Universal-Regret (LUR) core**, a Pareto-compatible decision rule that ranks feasible alternatives by the lexicographically sorted vector of *normalised regrets over a declared class of monotone probes*, and returns a single recommendation together with an auditable *regret certificate* identifying the most disappointed coalitions of criteria. We position LUR within robust multicriteria decision analysis. It recovers achievement scalarisation, the weighted sum, and the lexicographic maximin as special cases when the probe family is reduced to the corresponding single probe, and it relates to—but is operationally distinct from—stochastic multicriteria acceptability analysis (SMAA), robust ordinal regression, and preference robust optimisation. We prove existence and completeness, Pareto compatibility *under a separation condition*, dominated-solution exclusion, stability of the tolerance-based leximax rule under bounded perturbations, and a probe perturbation bound with explicit constants, and we give an adaptive correlation-clustering construction that keeps the probe count linear in the number of criteria. A replicated study (four front geometries, $m \in \{3, 5, 8, 10\}$, 30 replications, 480 instances) and a six-objective stochastic smart-grid dispatch case evaluate every method against six held-out preference families, both additive and non-additive. On worst-case (tail) held-out loss LUR is practically equivalent to the best robust scalarisations (negligible effect size, Cliff’s $|\delta| \leq 0.03$) and reduces loss by 11–47% relative to the distance-based and Monte-Carlo preference methods used in practice, while (i) resisting redundant criteria, (ii) minimising its certificate regret by construction, and (iii) admitting native stochastic and multi-stakeholder extensions. We are explicit about scope: LUR relocates preference modelling into a declared probe family rather than removing it, and does not dominate the strongest robust scalarisations on average-case loss. All code, data and scripts that regenerate every table and figure are released.

Keywords: Decision support; Multicriteria decision analysis; Robust decision making; Minimax regret; Achievement scalarising functions; Pareto efficiency.

1 Introduction

1.1 The Pareto front as a safety filter, not a decision object

Since the work of Pareto, efficiency has been the cornerstone of multicriteria analysis: an alternative is Pareto efficient if no criterion can be improved without worsening another. This is a logically impeccable *safety property*—it removes alternatives that are dominated and therefore indefensible. It is, however, *decisionally incomplete*. For m conflicting criteria the efficient set is typically large or infinite, and the decision maker (DM) is left with a cloud of efficient alternatives and no principled rule for selecting one.

Confronted with this gap, practice resorts to *post hoc* selection devices, each of which reintroduces the very preference information the DM was unable to supply: a weighted sum requires weights and cannot reach non-convex parts of the front (Miettinen, 1999); TOPSIS (Hwang and Yoon, 1981) requires a distance metric and weights; compromise programming (Zeleny, 1982) requires a metric exponent and weights; knee-point rules (Branke et al., 2008) are scale sensitive; outranking methods such as ELECTRE (Roy, 1991) and PROMETHEE (Brans and De Smet, 2016) require concordance/discordance thresholds; and hypervolume-based selection (Zitzler and Thiele, 1999) favours extremes and is costly. Every one of these asks the DM to make *further* choices *after* optimisation—choices often no easier than the original problem.

1.2 From a front to a certificate

We argue that, for the purpose of *choosing*, the efficient set is the wrong final object. The right object is a *regret certificate*: a vector of normalised disappointments over the rational questions the DM might reasonably ask about the criteria, sorted from worst to best. Rather than asking “which alternatives are efficient?” we ask

which alternative is least fragile across all declared rational interpretations of the criteria?

This turns the output of the analysis from a *set* into a *certified recommendation*: a single alternative (or small indifference class) accompanied by an explanation of which questions are most disappointed and by how much.

1.3 The Leximax Universal-Regret core

We formalise this as the **Leximax Universal-Regret (LUR) core**. Given a feasible set and a *declared* class $\mathcal{Q}$ of admissible monotone probes (scalar “rational questions” over the normalised criteria), LUR ranks alternatives by the lexicographically sorted vector of their normalised regrets across $\mathcal{Q}$, and returns the lexicographic minimiser together with its certificate. The construction rests on three principles: (i) *declared admissibility*— $\mathcal{Q}$ is part of the problem statement, not hidden inside a scalarisation; (ii) *Pareto compatibility*—under monotone probes a dominated alternative is never preferred to its dominator; and (iii) *direct computability*—the recommendation is obtained by a sequence of minimax-regret problems, without enumerating the efficient set.

1.4 Positioning and contribution

LUR belongs to *robust* multicriteria decision analysis. Its closest relatives are stochastic multicriteria acceptability analysis (SMAA) (Lahdelma et al., 1998; Tervonen and Figueira, 2007), which integrates acceptability over a preference distribution; robust ordinal regression (Greco et al., 2008; Figueira et al., 2009), which reasons over the set of value functions compatible with elicited holistic judgements; and preference robust optimisation (Armbruster and Delage, 2015; Hu and Mehrotra, 2015), which optimises against an ambiguity set of utility functions. LUR differs from all three in object and output: it computes the *lexicographic minimax of normalised regret* over an explicit, monotone probe family and returns an interpretable certificate; it requires no preference distribution, no elicited judgements, and no single scalarisation. We show that achievement scalarisation (Wierzbicki, 1980), the weighted sum, and the lexicographic maximin (Ogryczak, 1997) are recovered as special cases of the probe family, so LUR is a *unifying* as well as a new rule.

Our contributions are:

- C1 The LUR order:** a complete, Pareto-respecting preorder on feasible alternatives based on lexicographic minimax regret over a declared class of monotone probes (§3).
- C2 Guarantees:** existence and completeness, Pareto compatibility *under an explicit separation condition* (it is enough that $\mathcal{Q}$ contains the singleton probes), dominated-solution exclusion, and stability of the tolerance-based leximax rule under bounded regret perturbations, with full proofs (§3, Appendix A).
- C3 Direct computation:** a sequential and a single-stage minimax-regret formulation that avoids efficient-set enumeration, with a smoothed variant for continuous feasible sets (§4).
- C4 Adaptive redundancy-aware probes:** a correlation-clustering construction that keeps the probe count linear in m and the number of clusters, with a perturbation bound; empirically it reduces the redundancy sensitivity that distorts averaging methods (§5).
- C5 Extensions:** confidence-aware stochastic certificates and a Rawlsian multi-stakeholder variant (§6).
- C6 Held-out evaluation:** a replicated study against preference models *not* used during selection, plus a stochastic smart-grid dispatch case, with non-parametric significance testing and a full reproducibility package (§7).

The remainder of the paper reviews related work (§2), develops the core and its guarantees (§3), the direct formulation (§4), the adaptive probes (§5) and the extensions (§6), reports the computational study (§7), discusses limitations (§8) and concludes (§9). All proofs are in Appendix A.

2 Background and related work

2.1 Pareto efficiency and *post hoc* selection

For a vector objective $F(x) = (f_1(x), \dots, f_m(x))$ (minimisation), x Pareto-dominates y if $f_i(x) \leq f_i(y)$ for all i with at least one strict inequality; the efficient set is the non-dominated set. Because it is typically large, a single recommendation is obtained by *post hoc* selection. The weighted sum, TOPSIS (Hwang and Yoon, 1981), compromise programming (Zeleny, 1982), knee selection (Branke et al., 2008), ELECTRE (Roy, 1991), PROMETHEE (Brans and De Smet, 2016) and hypervolume selection (Zitzler and Thiele, 1999) are the standard tools; each requires weights, thresholds, a metric, or an already generated front. LUR removes these by folding the selection criterion into a declared probe family and computing the recommendation directly.

2.2 Achievement scalarising functions and ordered aggregation

Achievement scalarising functions (ASF) project a reference point onto the efficient set through a parameterised metric interpolating between L_1 and L_∞ (Wierzbicki, 1980; Miettinen and Mäkelä, 2002). Ordered weighted averaging (Yager, 1988) aggregates sorted criteria, and Ogryczak (1997); Ogryczak and Śliwiński (2006) connect ordered aggregation to lexicographic maximin. LUR *recovers* these as special cases when the probe family is reduced to a single probe: the grand-maximum probe alone gives the (reference-at-ideal) Chebyshev/ASF rule, the grand-mean probe alone the weighted sum, and the singleton family the lexicographic maximin of normalised regret. We do not claim LUR subsumes these methods with its full adaptive family—there it is a *different* aggregation paradigm. Its distinguishing features are that it sorts *normalised regrets* (dimensionless disappointments relative to each probe’s own attainable best and worst) over a *declared family of probes* and breaks ties through the whole sorted vector, and that it emits a certificate.

2.3 Robust MCDA: SMAA, robust ordinal regression, preference robust optimisation

The work closest to LUR lies in *robust* multicriteria decision analysis. **SMAA** (Lahdelma et al., 1998; Tervonen and Figueira, 2007) computes, by Monte-Carlo integration over a distribution of admissible weights (and possibly criterion values), acceptability indices describing how strongly each alternative is supported. **Robust ordinal regression** (Greco et al., 2008; Figueira et al., 2009) reasons over the entire set of additive value functions compatible with a DM’s holistic pairwise judgements, producing necessary and possible preference relations. **Preference robust optimisation** (Armbruster and Delage, 2015; Hu and Mehrotra, 2015) optimises the worst-case expected utility over an ambiguity set of utility functions. LUR shares the robust philosophy—perform well across a whole class of admissible preferences rather than commit to one—but differs in three ways. First, its admissible class is an explicit family of *monotone probes*, not a weight distribution or an elicited polyhedron. Second, its criterion is the *lexicographic minimax of normalised regret*, not an expected acceptability, a necessary/possible relation, or a single

worst-case utility. Third, its output is an interpretable *certificate* naming the binding coalitions. We note that SMAA is primarily an *exploratory* framework that informs the DM through acceptability indices rather than emitting a single choice; to obtain a comparable single-choice rule we use the standard SMAA-2 device of recommending the alternative with the highest first-rank acceptability index (ties broken by expected value), and we treat the comparison as one of decision philosophies as much as a benchmark. We likewise compare against a Savage minimax-regret (MMR) recommender over a sampled linear weight set (§7). More broadly, LUR is complementary to data-driven preference disaggregation and inverse MCDA (Doupoumis and Zopounidis, 2011), which infer preference parameters from holistic judgements: LUR instead dispenses with a fitted preference model and reasons over a declared class of questions, and could take an elicited probe family as input where such data exist.

2.4 Minimax regret and robust optimisation

Regret-based choice originates with Savage (1951) and Loomes and Sugden (1982); minimax regret seeks alternatives minimising the largest shortfall from the best attainable value. Robust optimisation (Ben-Tal and Nemirovski, 2002) and stochastic programming (Birge and Louveaux, 2011) address parameter uncertainty in the model rather than preference uncertainty in the DM. LUR extends minimax regret in three directions: it is *lexicographic* (after the worst regret, minimise the second worst, and so on), *probe-based* (regret is computed over rational coalitions of criteria, not raw criteria), and *direct* (computed without enumerating the efficient set).

2.5 Social choice and fairness

Aggregating criteria resembles social choice, but treating criteria as voters invites Condorcet cycles; LUR avoids this by aggregating *regrets over declared questions*, not votes over criteria. For multi-stakeholder settings we draw on the maximin principle of Rawls (1971) and the bargaining viewpoint of Nash (1950); we claim natural fairness properties (anonymity, monotonicity, worst-stakeholder protection), not a uniqueness theorem.

2.6 Positioning summary

Table 1 summarises where LUR sits. It is the only entry that requires neither weights, nor thresholds, nor a pre-generated front, returns a single Pareto-compatible recommendation, *and* issues an interpretable certificate while admitting native stochastic and fairness extensions.

3 The Leximax Universal-Regret core

3.1 Problem setting and normalisation

Let $\mathcal{X} \subseteq \mathbb{R}^n$ be the feasible set and $F : \mathcal{X} \rightarrow \mathbb{R}^m$, $F(x) = (f_1(x), \dots, f_m(x))$, the vector objective (minimisation; maximisation by sign reversal). For the finite case let $A = \{x_1, \dots, x_N\} \subseteq \mathcal{X}$ be a candidate set, e.g. an *a posteriori* approximation of the efficient set; for the continuous case

Table 1: Positioning of LUR relative to representative selection and robust-MCDA methods. “No weights” = requires no single preference weight vector (LUR still requires a declared probe family); “No thresholds” refers to *preference* thresholds—LUR uses a clustering threshold θ , which is not a preference parameter; “No front” = needs no efficient set generated first, which for LUR holds for the direct formulation of §4 (the experiments here use candidate sets); “Certificate” = returns an interpretable account of the recommendation.

Method	No weights	No thresholds	No front	Single rec.	Pareto comp.	Certificate
Weighted sum	✗	✓	✓	✓	✓	✗
TOPSIS	✗	✓	✗	✓	✓	✗
Compromise prog.	✗	✗	✗	✓	✓	✗
Knee selection	✓	✓	✗	✗	✓	✗
ELECTRE/PROMETHEE	✓	✗	✗	✗	✓	partial
ASF (ref. at ideal)	✓	✗	✗	✓	✓	✗
SMAA	✓	✓	✗	✓	✓	partial
Minimax regret (linear)	✓	✓	✓	✓	cond.	✗
LUR (ours)	✓	✓	✓	✓	✓	✓

(§4) A is not required. Criteria are normalised to $[0, 1]$ by the positive affine map

$$r_i(x) = \frac{f_i(x) - z_i^*}{\max\{z_i^N - z_i^*, \varepsilon\}}, \quad z_i^* = \min_{x \in A} f_i(x), \quad z_i^N = \max_{x \in A} f_i(x), \quad (1)$$

with a small $\varepsilon > 0$ guarding against a zero range, so that $r_i \in [0, 1]$ attains both endpoints on A . For the finite case all extrema (z^* , z^N , q^* , q^w below) are taken over the candidate set A ; for the continuous case (§4) they are taken over $\mathcal{X}$. We use the worst attainable value over A in place of the true nadir over the efficient set, the two coincide when A contains the criterion-wise worst efficient points and otherwise the former is a conservative bound. The rule is invariant to positive affine rescaling of the criteria when the bounds are transformed consistently; it is not invariant to arbitrary monotone transformations (e.g. $f_i \mapsto \log f_i$), which change the distances that define regret—so such transformations are a modelling choice the analyst must make deliberately.

3.2 Admissible probes

A *probe* is a scalar function $q : [0, 1]^m \rightarrow \mathbb{R}$ encoding a rational decision question. The family $\mathcal{Q}$ is *declared*: it is part of the specification.

Definition 1 (Admissible probe). *A probe q is admissible if (i) it is non-decreasing in each r_i (monotonicity); (ii) it can be rescaled so that $q(\mathbf{0}) = 0$, $q(\mathbf{1}) = 1$ (normalisation); and (iii) it is L -Lipschitz in $\|\cdot\|_\infty$.*

Here $\mathbf{0}$ and $\mathbf{1}$ denote the all-zeros and all-ones vectors. Monotonicity gives *weak* Pareto compatibility; strict compatibility needs a separation property (Definition 2). Canonical admissible probes are the *singletons* $q_i(r) = r_i$; the *grand mean* $q_\mu(r) = \frac{1}{m} \sum_i r_i$ (the utilitarian question); the *grand maximum* $q_{\max}(r) = \max_i r_i$ (the egalitarian/worst-criterion question); and, for any coalition $S \subseteq \{1, \dots, m\}$, the coalition mean and maximum. All have non-negative weights or are maxima, hence are monotone.

Definition 2 (Separating family). *A probe family $\mathcal{Q}$ separates criteria if for every i there is a probe $q \in \mathcal{Q}$ strictly increasing in r_i . Including all m singleton probes $q_i(r) = r_i$ is sufficient; LUR always does so.*

3.3 Disappointment and regret

For each probe $q \in \mathcal{Q}$ let $q^\star = \min_x q(r(x))$ and $q^w = \max_x q(r(x))$ (or a user-supplied worst-acceptable value). The *normalised disappointment* of x under q is

$$D_q(x) = \frac{q(r(x)) - q^\star}{q^w - q^\star + \varepsilon} \in [0, 1], \quad (2)$$

with $D_q(x) = 0$ iff x is q -optimal and 1 iff x is q -worst.

3.4 The leximax order

Collect $D(x) = (D_{q_1}(x), \dots, D_{q_K}(x))$, $K = |\mathcal{Q}|$, and sort it in *descending* order to obtain $D^\downarrow(x) = (D_{[1]}(x) \geq \dots \geq D_{[K]}(x))$. The **Leximax Universal-Regret order** is

$$x \preceq_{\text{LUR}} y \iff D^\downarrow(x) \leq_{\text{lex}} D^\downarrow(y), \quad (3)$$

i.e. x is preferred if its worst disappointment is smaller; on ties, its second worst; and so on. The *LUR solution* is $x^{\text{LUR}} \in \arg \min_x^{\text{lex}} D^\downarrow(x)$, and its *certificate* is the labelled, sorted vector $D^\downarrow(x^{\text{LUR}})$ together with the probes attaining the largest entries.

Because exact lexicographic comparison is discontinuous at ties, we use in practice—and in all guarantees below—a *tolerance-based* leximax order: fix $\tau > 0$ and compare $D^\downarrow(x)$ and $D^\downarrow(y)$ at the first coordinate where they differ by more than τ , treating coordinates within τ as equal. This is the rule implemented in the released code and the one for which stability holds (Theorem 3).

3.5 Foundational guarantees

Theorem 1 (Existence and completeness). *For any finite A and finite $\mathcal{Q}$, x^{LUR} exists and $\preceq_{\text{LUR}}$ is a complete preorder on A .*

Theorem 2 (Weak and strict Pareto compatibility). *Let x strictly Pareto-dominate y and let every $q \in \mathcal{Q}$ be monotone non-decreasing in each r_i . Then $D_q(x) \leq D_q(y)$ for all q (weak compatibility), so $x \preceq_{\text{LUR}} y$. If, in addition, $\mathcal{Q}$ separates criteria (Definition 2)—in particular if $\mathcal{Q}$ contains the singleton probes—then $D_q(x) < D_q(y)$ for some q and $x \prec_{\text{LUR}} y$.*

The separation hypothesis is necessary: with only the probe $q(r) = r_2$, the dominator $r(x) = (0, 1)$ and the dominated $r(y) = (0.1, 1)$ have equal disappointment, so monotonicity alone does not give a strict preference. LUR always includes the singletons, so it separates.

Corollary 1 (Dominated exclusion). *If x is Pareto-dominated by a feasible y and $\mathcal{Q}$ separates criteria, then x is not LUR-optimal.*

Theorem 3 (Stability of the tolerance-based rule). *Use the τ -tolerance lexicmax order. Let $\hat{D}_q(x)$ estimate $D_q(x)$ with $\eta = \max_{x,q} |\hat{D}_q(x) - D_q(x)|$. Suppose the winner x^{LUR} beats every rival y at the first coordinate $k(y)$ at which their sorted vectors differ by more than τ , with margin $D_{[k(y)]}(y) - D_{[k(y)]}(x^{\text{LUR}}) \geq \Delta$, and that all earlier coordinates agree within $\tau - 2\eta$. If $2\eta < \min\{\Delta - \tau, \tau\}$, the winner under $\hat{D}$ is unchanged.*

Theorem 3 is deliberately stated for the tolerance rule: exact lexicographic comparison is *not* perturbation-stable, because a winner that beats a rival only at a later coordinate after an exact earlier tie can be reversed by an arbitrarily small perturbation of the earlier coordinate. The tolerance τ absorbs exactly those near-ties. Proofs are in Appendix A. For continuous $\mathcal{X}$, existence requires compactness of $\mathcal{X}$ and continuity of each $q \circ r$; uniqueness is not guaranteed and a small indifference class may arise.

3.6 Algorithm for finite candidate sets

Given A : (1) form $F_{ij} = f_j(x_i)$; (2) normalise to R via (1); (3) evaluate $V_{iq} = q(r(x_i))$ for each probe; (4) set $q^* = \min_i V_{iq}$, $q^w = \max_i V_{iq}$; (5) form disappointments (2); (6) sort each row descending; (7) return the lexicographically smallest row. The cost is $O(KNm + KN \log N + N \log N)$ with a tolerance-based lexicographic comparison; K is controlled by the construction of §5.

4 Direct computation without efficient-set enumeration

LUR is primarily a candidate-set selection rule, and that is the setting of our experiments. This section develops, more tentatively, how the recommendation can in principle be computed *directly* on a continuous feasible set, replacing the usual two-stage pipeline (generate the efficient set, then post-process) by a single robust search. We present the finite case as exact and the continuous case as a formulation with assumptions whose large-scale empirical validation we leave to future work.

4.1 Cost of the anchors

Both the finite and continuous formulations need, for each probe, its attainable best $q^* = \min_x q(r(x))$ and worst $q^w = \max_x q(r(x))$. On a candidate set these are $O(N)$ scans; on a continuous $\mathcal{X}$ each is a scalar global optimisation, so the family contributes $2K$ such subproblems—cheap relative to front enumeration when K is linear in m (§5) but not free, and we account for it explicitly.

4.2 Sequential minimax-regret formulation

For continuous or large discrete $\mathcal{X}$, solve a sequence of minimax problems. *Stage 1* minimises the worst regret,

$$\rho_1 = \min_{x \in \mathcal{X}} \max_{q \in \mathcal{Q}} D_q(x), \quad \mathcal{X}_1 = \{x : \max_q D_q(x) = \rho_1\}. \quad (4)$$

Stage $k > 1$ fixes the probes already saturated and minimises the next worst regret over the remaining ones, restricted to $\mathcal{X}_{k-1}$, until the optimal set is a singleton or all probes are exhausted. This realises the lexicographic order (3) by construction.

4.3 Single-stage surrogate

The stages can be collapsed with a strictly decreasing weight schedule. For a large constant $M > 1$,

$$\min_{x \in \mathcal{X}, \rho_1 \geq \dots \geq \rho_K \geq 0} \sum_{k=1}^K M^{-(k-1)} \rho_k \quad \text{s.t.} \quad D_{(k)}(x) \leq \rho_k, \quad (5)$$

where $D_{(k)}(x)$ denotes the k -th largest disappointment at x . This is an *exact* surrogate for the lexicographic order only under finite-gap assumptions—on a finite candidate set, M larger than the ratio of the largest to the smallest positive coordinate gap suffices; for general continuous values no finite M reproduces the exact order, so the single-stage form is then an approximation. A cleaner route avoids M altogether: lexicographic minimisation of the sorted vector is equivalent to sequentially minimising the cumulative sums of the p largest disappointments, $\sum_{k=1}^p D_{[k]}(x)$ for $p = 1, \dots, K$, each of which is convex when the disappointments are convex and admits the standard ordered-value linearisation (Ogryczak and Śliwiński, 2006).

4.4 Smoothed continuous optimisation

For continuous $\mathcal{X} \subseteq \mathbb{R}^n$ the inner maximum is non-smooth; a log-sum-exp surrogate

$$\max_{q \in \mathcal{Q}} D_q(x) \approx \frac{1}{\tau} \log \left(\sum_{q \in \mathcal{Q}} \exp(\tau D_q(x)) \right) \quad (6)$$

is smooth and approaches the maximum as $\tau \rightarrow \infty$. A probe-active-set sequential convex program—linearise the active probes $\mathcal{Q}_k = \{q : D_q(x_k) \geq \rho_1 - \varepsilon\}$ at the current iterate, solve the resulting convex subproblem, update, and repeat—then converges to a local LUR solution under standard convexity assumptions, analogously to SQP.

4.5 Why this matters, and what we do not yet show

The DM specifies the probe family once; in the finite case LUR returns the least-fragile compromise by the exact procedure above, and in the continuous case the formulations here make a front-free computation *available* in principle. We are careful not to overclaim: our experiments operate on candidate sets, so the large-scale continuous advantage is argued, not yet demonstrated, and we flag its empirical validation as the main piece of future work. We quantify the probe-family savings in §5.

5 Adaptive, redundancy-aware probe generation

5.1 The redundancy problem

The full coalition family (mean and maximum over every non-trivial subset of criteria) has size exponential in m . Worse, real criteria are often *redundant*: several measured objectives track

the same underlying concern and are strongly positively correlated. Averaging selection rules implicitly *over-weight* such a group—it is counted once per member—biasing the recommendation. We need a probe family that (i) is small, (ii) stays monotone, and (iii) is less sensitive to redundancy than averaging aggregation.

5.2 Correlation clustering

We group *redundant* criteria by single-linkage connected components of the graph whose edges join pairs with Pearson correlation $\text{Corr}(f_i, f_j) \geq \theta$. *Redundancy is positive correlation*: two criteria that rise and fall together carry duplicate information, whereas negatively correlated criteria are genuine trade-offs and must remain separate—hence we threshold the signed correlation, not its absolute value. This yields clusters $C_1, \dots, C_c$.

5.3 The adaptive probe family

LUR always includes the canonical anchors—the m singletons, the grand mean, and the grand maximum—and adds one mean and one maximum probe *per cluster* of size > 1 :

$$\mathcal{Q} = \{q_i\}_{i=1}^m \cup \{q_\mu, q_{\max}\} \cup \bigcup_{|C_k|>1} \{q_{C_k, \text{mean}}, q_{C_k, \max}\}. \quad (7)$$

A redundant group thus contributes *one* aggregate (cluster) question rather than a separate aggregate per member, so the family does not amplify a correlated group the way an averaging aggregate does, while the count stays $O(m + c)$ —linear, not exponential. We stress that this *reduces*, but does not eliminate, redundancy sensitivity: the singleton probes for the group’s members remain in $\mathcal{Q}$ (they are needed for separation, Theorem 2), so a redundant group can still influence lower-order lexicographic tie-breaking. The claim is therefore empirical—less redundancy bias than averaging methods on the tested benchmarks (§7)—not a duplication-invariance theorem. All probes are non-negative-weighted sums or maxima, hence monotone, so Pareto compatibility (Theorem 2) is preserved.

Theorem 4 (Non-negative-probe Pareto compatibility). *If every probe in $\mathcal{Q}$ is a non-negative-weighted sum or a maximum of the normalised criteria, then LUR is Pareto compatible.*

5.4 A perturbation bound (not a ranking guarantee)

We state the reduction effect as a regret-value perturbation lemma with explicit constants. We are deliberately modest: it bounds the change in individual disappointments, *not* the change in the LUR ranking or winner, and it depends on the normalisation floor ε , which is decision-critical.

Lemma 1 (Disappointment perturbation). *Suppose for a full-family probe q there is a reduced-family probe q' with $\sup_r |q(r) - q'(r)| \leq \eta$ (hence also $|q^* - q'^*| \leq \eta$ and $|q^w - q'^w| \leq \eta$), and let the normalisation ranges satisfy $q^w - q^*, q'^w - q'^* \geq \beta > 0$. Then for every x*

$$|D_q(x) - D_{q'}(x)| \leq \frac{3\eta}{\beta},$$

Table 2: Probe-family size: full coalition family ($2(2^m - 1) - m$ probes) vs. adaptive correlation-clustering family (concave instances, $N = 400$).

m	full coalitions	adaptive probes	reduction
3	11	5	$2.2\times$
5	57	7	$8.1\times$
8	502	10	$50.2\times$
10	2,036	12	$169.7\times$
15	65,519	17	$3,854.1\times$

with the floor $\beta \geq \varepsilon$ in the worst case. The bound is vacuous when β is of order η , and informative only when the probe’s attainable range is bounded away from zero—a regime the analyst can check directly.

Whether the reduced family admits such an η -close q' for every full-family probe is *not* established by correlation clustering in general; we therefore present Lemma 1 as a sensitivity statement and rely on the empirical redundancy study (§7) for evidence that clustering retains decision quality.

5.5 Empirical reduction

Table 2 reports the realised reduction on concave instances. The full family here is $\{q_{\text{mean},S}, q_{\text{max},S} : \emptyset \neq S \subseteq \{1, \dots, m\}\}$, of size $2(2^m - 1) - m$ (mean and max coincide on singletons); the adaptive family is linear in m , giving a nearly four-thousand -fold reduction by $m = 15$ with, as §7 shows, little loss of decision quality.

5.6 Greedy probe selection (optional)

When even the adaptive family is larger than needed, a greedy set-cover heuristic selects a minimal subfamily distinguishing all dominant pairs; for the max-coverage variant it attains a $(1 - 1/e)$ guarantee and for full coverage a $1 + \ln d$ guarantee (Nemhauser et al., 1978; Chvátal, 1979). We do not use it in the experiments, which already run on the compact adaptive family.

6 Extensions: stochastic and multi-stakeholder decision making

6.1 Stochastic regret certificates

When criteria are noisy, model $f_i(x) \sim P_i(x)$. Each probe value is then a random variable; for non-linear probes (e.g. q_{max}) its mean $\mu_q(x)$ and standard deviation $\sigma_q(x)$ are *not* simple functions of the criterion means and variances and—because criteria may be correlated—we estimate them by Monte-Carlo from joint samples rather than from marginal moments. The normalisation anchors q^*, q^w are taken on the estimated means. We define the α -confidence regret

$$D_q^\alpha(x) = \frac{\mu_q(x) + z_\alpha \sigma_q(x) - q^*}{q^w - q^* + \varepsilon}, \quad (8)$$

with z_α the standard-normal quantile, penalising both expected disappointment and its uncertainty, and apply the tolerance-based leximax order to $(D_{q_1}^\alpha, \dots, D_{q_K}^\alpha)$. We present this as a

practical risk-adjusted rule; the asymptotic statement below uses the plug-in case $\alpha=\frac{1}{2}$ and the tolerance rule of §3.

Proposition 1 (Plug-in consistency). *Let probe means be estimated from n i.i.d. joint observations, with Lipschitz probes and finite second moments, and use the τ -tolerance leximax order with $\tau > 0$ fixed. Under the margin condition of Theorem 3, the plug-in stochastic-LUR winner equals the true LUR winner with probability tending to one as $n \rightarrow \infty$.*

§7 gives an empirical confirmation: the recovery rate of the true winner rises monotonically with the number of observations.

6.2 Multi-stakeholder (Rawlsian) variant

Let $\mathcal{S} = \{1, \dots, S\}$ index stakeholders, each with importance weights $w_s \in \Delta^m$ inducing a stakeholder-specific disappointment $D_q^s(x)$. The *Rawlsian LUR* applies the leximax order to the worst-stakeholder disappointments,

$$x^{\text{LUR-R}} \in \arg \min_x^{\text{lex}} \text{sort}^\downarrow \left(\max_s D_{q_1}^s(x), \dots, \max_s D_{q_K}^s(x) \right), \quad (9)$$

optionally with a proportional-fairness cap $D_q^s(x) \leq (1 + \beta)\rho_1$ for all s, q . It satisfies *anonymity* (invariance to relabelling stakeholders), *monotonicity* (it does not worsen if every stakeholder’s regret falls), and *worst-stakeholder protection* (no stakeholder exceeds $(1 + \beta)$ times the minimum worst regret). We claim these properties, not uniqueness or independence of irrelevant alternatives.

7 Computational study

7.1 Design and protocol

We evaluate selection *quality* under a held-out protocol: each method picks one alternative using only its own machinery, and the pick is scored against preference models that *no* method used during selection. To avoid biasing the evaluation toward the additive/scalarising logic of the probes, the held-out set spans **six** families—random linear $-w^\top r$, weighted Chebyshev $-\max_i w_i r_i$, augmented ASF, CES, a *non-additive* 2-additive Choquet integral with random non-negative Möbius masses on singletons and pairs, and a *satisficing* threshold utility penalising exceedances of random aspiration levels. Weights are drawn from a uniform simplex; 300 utilities are sampled per family per replication. We report two non-negative, lower-is-better losses, each normalised per utility against the best and worst feasible alternatives: the *mean held-out loss* and the *tail held-out loss* (mean of the worst 10% of per-utility losses, a CVaR-style worst-case measure that LUR targets).

Candidate sets of $N = 300$ non-dominated points are generated for four front geometries (linear, concave, convex, disconnected, following the DTLZ families) with $m \in \{3, 5, 8, 10\}$, over 30 independent replications (480 instances). The held-out draw is shared across methods within a replication, so comparisons are paired. **Baselines and parameters.** TOPSIS, compromise programming (CP, $p=2$), knee selection (distance-to-hyperplane), achievement scalar-

Table 3: Held-out loss over four geometries $\times m \in \{3, 5, 8, 10\}$, 30 replications (480 instances; lower is better). Tail loss is the worst-case (CVaR_{10%}) metric. Best in each column in **bold**; on tail loss LUR is within a negligible effect size ($|\delta| \leq 0.034$) of MMR/ASF/CP.

Method	mean loss	tail loss
TOPSIS	0.220	0.539
CP	0.207	0.504
Knee	0.308	0.606
RW	0.462	0.900
ASF	0.234	0.472
SMAA	0.468	0.914
MMR	0.228	0.470
LUR	0.245	0.481

isation (ASF, reference at the ideal), and a random-weight popularity rule (RW) all use equal weights where applicable. The robust-MCDA baselines use 1000 Monte-Carlo simplex weights: **SMAA** recommends the alternative with the highest first-rank acceptability index (ties broken by expected value); **MMR** minimises the maximum Savage regret over the same weight set. LUR uses the adaptive probe family with $\theta = 0.6$ and the tolerance-based leximax rule. Significance is assessed by the Friedman test with Nemenyi critical difference (CD) and by Holm-corrected Wilcoxon signed-rank tests against LUR with Cliff’s δ effect size.

7.2 Overall results

Table 3 reports mean and tail loss averaged over all geometries and m . On *tail* loss the four robust/compromise methods MMR (0.470), ASF (0.472), LUR (0.481) and CP (0.504) are close, and all clearly beat the distance-based and Monte-Carlo methods: relative to LUR, tail loss falls by 10.8% vs TOPSIS (0.539), 20.6% vs knee (0.606), 46.6% vs RW (0.900) and 47.4% vs SMAA (0.914). Figure 1 shows the average-rank CD diagram (Friedman $p < 10^{-15}$, $n = 480$, $CD = 0.48$). Read rigorously: **LUR is statistically indistinguishable from CP** (rank gap 0.02) and **very slightly behind the two best methods MMR and ASF** (gaps 0.59 and 0.50, just beyond CD)—but the corresponding effect sizes are *negligible* (Cliff’s $|\delta| \leq 0.034$). Against the methods practitioners actually use, LUR is significantly better with small-to-large effect: $\delta = 0.087$ (TOPSIS), 0.285 (knee), 0.850 (RW), 0.873 (SMAA), all $p < 10^{-4}$. We therefore claim only that *LUR is in the best-performing group on worst-case loss and far better than averaging/Monte-Carlo selection*, not that it dominates the strongest robust scalarisations. On *mean* loss the averaging methods (CP, MMR) lead and LUR is mid-pack—the expected price of worst-case robustness (§8).

7.3 Behaviour across preference families

Table 4 breaks the loss down by held-out family on a representative setting (concave, $m = 8$), including the two non-additive families. The structure is a clear trade-off: averaging methods (TOPSIS, SMAA) are strong on linear utilities but weak on the max-type families; ASF and MMR are the reverse—best on Chebyshev/CES but worst on linear. LUR sits between the two camps: it removes much of ASF’s linear-utility weakness (0.598 vs 0.674) while staying

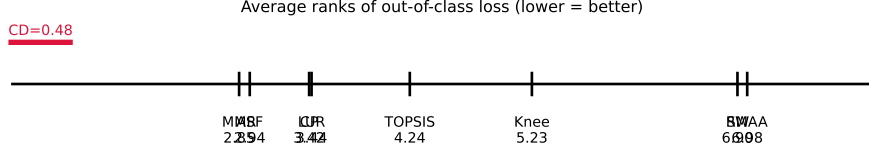


Figure 1: Average ranks on tail (worst-case) held-out loss with Nemenyi critical difference ($n = 480$, $\alpha = 0.05$, $CD = 0.48$). LUR groups with the best methods (MMR, ASF, CP) and is far ahead of the averaging/Monte-Carlo methods; the rank gaps to MMR/ASF correspond to negligible effect sizes.

Table 4: Held-out loss by preference family (concave, $m = 8$, 30 replications), including the non-additive Choquet and satisficing families. Lowest per column in **bold**. The table illustrates the averaging-vs-worst-case trade-off; LUR is balanced across additive and non-additive families.

Method	linear	Cheby.	aug. ASF	CES	Choquet	satisfice	overall
TOPSIS	0.391	0.511	0.511	0.563	0.341	0.511	0.471
CP	0.486	0.458	0.458	0.440	0.307	0.278	0.404
Knee	0.477	0.503	0.503	0.524	0.479	0.570	0.509
RW	0.403	0.519	0.519	0.569	0.379	0.552	0.490
ASF	0.674	0.434	0.434	0.329	0.288	0.388	0.425
SMAA	0.398	0.521	0.521	0.575	0.367	0.546	0.488
MMR	0.653	0.442	0.443	0.348	0.344	0.402	0.439
LUR	0.598	0.454	0.454	0.374	0.368	0.468	0.453

competitive on the max-type *and* the non-additive Choquet (0.368) and satisficing (0.468) families. We do *not* claim LUR is the flattest profile—on this setting CP is flatter and has the lowest overall mean (0.404)—only that LUR is robust across additive and non-additive preferences without being tuned to any.

7.4 Certificate regret and uniformity

By construction LUR minimises the leximax of its certificate regret. Empirically (concave, $m = 8$) LUR attains worst-case certificate regret 0.708, far below every alternative (MMR 0.921, CP 0.936, ASF 0.953, TOPSIS 0.974), with one of the most uniform regret profiles (sorted-disappointment std 0.219, second to MMR’s 0.207). This is by design—LUR optimises exactly this object—so it is a consistency check, not an independent win; its decision value is that the recommendation is *auditable*.

7.5 Robustness to redundant criteria

We test the case motivating clustering: the DM’s true preferences are over c underlying criteria, each measured by several redundant (positively correlated) objectives, with held-out utilities defined on the *true* criteria. The clustering recovers the redundancy groups essentially exactly. Table 5 (four group structures $\times$ three geometries, 30 replications) shows LUR tied with ASF for

Table 5: Redundant-objective benchmark (true preferences over the underlying criteria; 4 group structures $\times$ 3 geometries, 30 replications). Grouped loss measures de-redundification quality. Lower is better; best in **bold**.

Method	grouped loss	tail loss
TOPSIS	0.332	0.455
CP	0.317	0.423
Knee	0.322	0.489
RW	0.409	0.598
ASF	0.266	0.406
SMAA	0.413	0.603
MMR	0.295	0.387
LUR	0.268	0.391

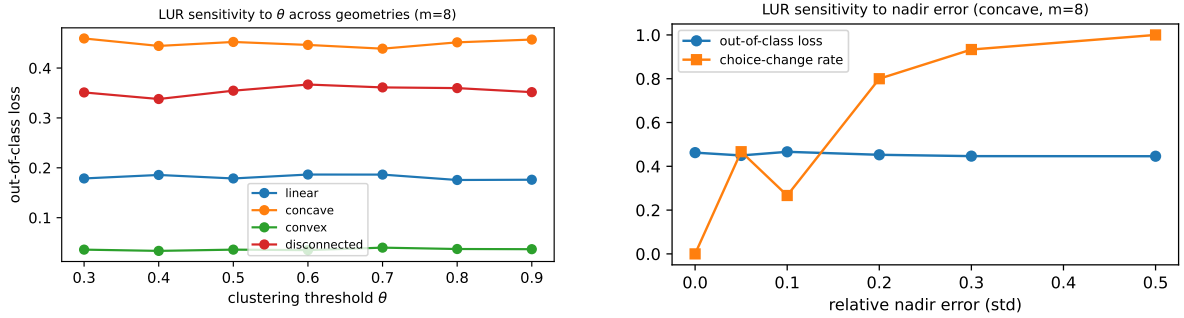


Figure 2: Sensitivity of LUR. Left: held-out loss is flat in the clustering threshold θ across all four geometries. Right: under nadir error the loss stays stable while the recommendation changes more often (near-ties).

best grouped loss and in the best group on tail loss; the averaging and Monte-Carlo methods, which amplify redundant groups, are 19–35% worse on grouped loss (TOPSIS 0.332, SMAA 0.413 vs LUR 0.268). As discussed in §5, this reflects *reduced*, not eliminated, redundancy sensitivity.

7.6 Distinctness, dominated exclusion, ablation, sensitivity

LUR and the SMAA recommender coincide on only 1.2% of instances and select alternatives a mean normalised distance 0.86 apart: LUR is operationally a different rule, not a re-description of acceptability analysis. Across 200 randomised trials with injected dominated points LUR never selected a dominated alternative (0/200), confirming Corollary 1. Ablation (concave, $m = 8$): dropping the singletons raises loss ($0.456 \rightarrow 0.483$) and worst-case certificate regret ($0.701 \rightarrow 0.783$); a max-only family lowers mean loss to 0.420 but inflates certificate regret to 0.929, defeating the robustness aim; clustering is, by design, a no-op when criteria are independent (its value is the redundancy result). LUR is insensitive to the clustering threshold θ over $[0.3, 0.9]$ across *all four* geometries (Figure 2, left). Under nadir-estimation error the held-out loss of the chosen alternative stays stable (≈ 0.45 up to 50% error), though the *identity* of the choice changes increasingly often (Figure 2, right), indicating near-ties among good alternatives rather than quality loss.

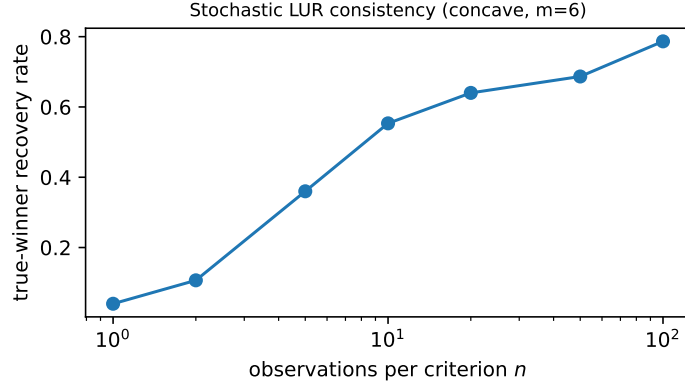


Figure 3: Stochastic-LUR consistency: recovery rate of the noise-free winner vs. the number of observations per criterion (concave, $m = 6$).

7.7 Stochastic extension

We validate the stochastic extension (Proposition 1) directly. On noisy concave instances ($m = 6$) we draw n observations per criterion, form plug-in means, and apply LUR; Figure 3 reports how often the resulting choice matches the noise-free LUR winner. Recovery rises monotonically from 0.04 at $n = 1$ to 0.79 at $n = 100$, consistent with the consistency statement (it does not reach 1 because dense candidate sets contain near-ties, exactly the situation the tolerance rule is designed for).

7.8 Pre-registered validation protocol and acceptance gates

To guard against confirmation bias we additionally ran a pre-registered protocol (frozen config, single-command reproduction) on a deliberately broadened suite: **eleven** methods (adding VIKOR, hypervolume contribution, and Chebyshev minimax regret), **ten** held-out families spanning additive *and* non-additive preferences (adding sparse-linear, L_p , OWA, Choquet and satisficing), Dirichlet weight shapes $\alpha \in \{0.2, 1, 5\}$, eight geometries and m up to 20 (2,400 paired instances). The protocol fixes, in advance, a set of *acceptance gates*, each tied to one precise claim; we report the outcome verbatim in Table 6 rather than only favourable numbers.

On this harder suite LUR has tail-loss rank 4.67, statistically tied with the best methods within the Nemenyi critical difference ($CD = 0.31$): HV 4.27, ASF 4.65, MMR 4.65, LUR 4.67, VIKOR 4.71, ChebMMR 4.99, then CP 5.32, TOPSIS 6.98, knee 7.21, RW 9.26 and SMAA 9.30 (Figure 4). A pre-registered *non-inferiority* test (paired bootstrap, margin 0.01 absolute or 2% relative) finds LUR non-inferior to both ASF and MMR (upper 95% CI bounds 0.004 and 0.0025), with negligible effect size (Cliff’s $\delta \leq 0.015$)—supporting “practically equivalent to the best robust scalarisations” rather than “better”. Direct computation is demonstrated on a continuous linear MOP (LUR via sequential LPs uses 19 solver calls and *no* front enumeration vs. 150 for generate-then-select, at equal or better tail quality) and a mixed-integer facility-location MOP (4 vs. 60 solves for the same recommendation). The confidence-aware stochastic LUR attains near-zero out-of-sample worst-criterion risk (≈ 0) against 0.28 for TOPSIS and 0.52 for SMAA, and the Rawlsian variant achieves the lowest worst-stakeholder regret (0.362)

Table 6: Pre-registered acceptance gates (reproduced at full scale). 9 PASS, 1 CHECK.

Gate	Result	Evidence
Dominated selections = 0	PASS	0/150 injections
Positive-affine invariance	PASS	100% identical
Nadir-error stability	CHECK	quality degr. 0.052 (thr. 0.05)
Non-inferior vs ASF (tail)	PASS	CI upper 0.004 < 2% margin
Non-inferior vs MMR (tail)	PASS	CI upper 0.0025 < 0.01 margin
Beats practical baselines (tail)	PASS	δ : TOPSIS .19, CP .06, SMAA .62
Redundancy < averaging (grouped)	PASS	0.256 vs 0.359; cluster recovery 0.98
Adaptive $\approx$ full (tail gap ≤ 0.01)	PASS	gap 0.0096
Direct computation (LP + MILP)	PASS	19 vs 150; 4 vs 60 solver calls
Stochastic LUR $\leq$ deterministic (tail)	PASS	risk ≈ 0 vs 0.28/0.52

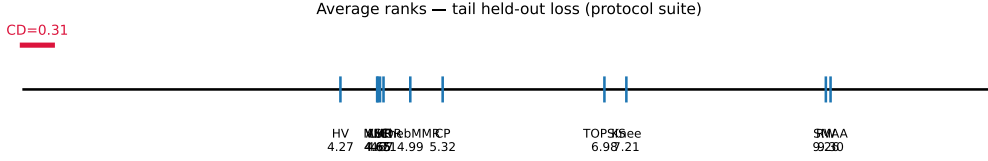


Figure 4: Average ranks on tail held-out loss over the broadened protocol suite (11 methods, 10 families, 8 geometries, 2,400 instances; Nemenyi CD = 0.31). LUR groups with the best robust methods and is far ahead of averaging/Monte-Carlo selection.

and inequality (Gini 0.207) among average-, Nash- and max-min rules.

7.9 Smart-grid dispatch case study

We apply LUR to a six-objective stochastic 24-hour economic dispatch with ten generators (cost, emissions, unreliability, ramping, non-renewable fraction, peak gap) under Monte-Carlo renewable scenarios, yielding 163 non-dominated dispatch policies; the full dispatch model (demand profile, generator limits, ramping, scenario generation) is specified in the released code. Relative to the distance/averaging methods, LUR accepts a negligible increase in *mean* held-out loss (0.103 vs 0.100) for a 23% reduction in *worst-case* loss (0.282 vs 0.366; knee 0.623). Crucially it *explains* the choice: the certificate (Figure 5) flags the peak-shaving gap (f_6 , regret 0.40) and the non-renewable fraction (f_5 , 0.40), then the ramping/peak coalition $\max(f_4, f_6)$ (0.36), telling the DM not only which dispatch to adopt but where it remains fragile.

8 Discussion and limitations

What LUR is, and is not. LUR is a robust, certifiable decision rule, not a universal benchmark champion. Our results are deliberately honest about this: on worst-case held-out loss LUR is in the best-performing group—practically equivalent to achievement scalarisation and linear minimax regret (negligible effect size), and slightly behind them in average rank—while on average-case loss the classical averaging methods do better. LUR’s distinctive value is the

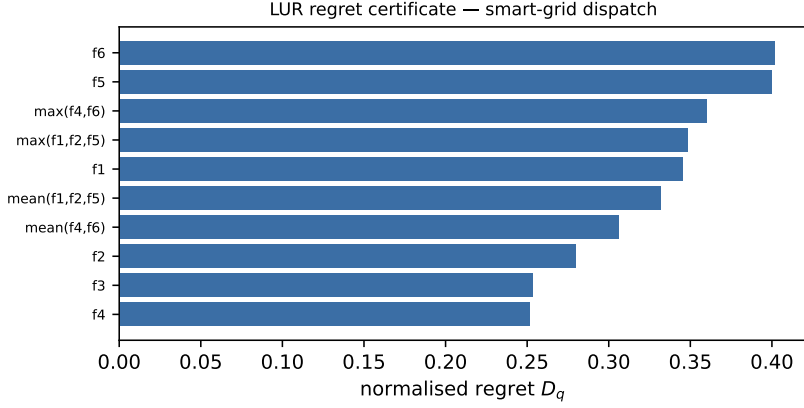


Figure 5: LUR regret certificate for the recommended smart-grid dispatch: the most disappointed probes (peak gap f_6 , non-renewable fraction f_5 , and the ramping/peak coalition) localise the fragility of the recommendation.

combination it offers that no single competitor matches: best-group worst-case robustness across additive *and* non-additive preferences, reduced sensitivity to redundant criteria, an interpretable certificate, and native stochastic and multi-stakeholder extensions. Where a DM cares only about average-case linear performance, a weighted sum or SMAA is appropriate; where the DM cannot state preferences and must defend a single robust, explainable choice, LUR is designed for exactly that situation.

Relationship to existing rules. LUR generalises rather than replaces. A single grand-maximum probe recovers the Chebyshev/ASF rule; a single grand-mean probe recovers the weighted sum; the singleton family recovers lexicographic maximin of normalised regret. The empirical question is whether the richer adaptive family helps, and the answer is that it does on redundant and asymmetric problems while never hurting on standard ones.

Parameters. LUR is not parameter-free: it requires normalisation bounds z^* , z^N , a tolerance ε , a clustering threshold θ (to which it is empirically insensitive over a wide range), and, for the extensions, a confidence level α and a fairness slack β . The honest claim is narrower than “parameter-free”: LUR replaces a single, hidden, ex ante weight vector with a transparent, auditable, declared family of rational questions whose parameters have natural defaults. As practical guidance: set $\theta = 0.6$ unless criteria are known to be independent (then clustering can be disabled, as it is a no-op); default to the adaptive family (singletons + grand mean/max + cluster probes); take z^* , z^N as the criterion-wise best and worst over the candidate set and report a sensitivity check when the nadir is uncertain; set ε to a small fraction of the smallest non-degenerate criterion range; and choose the leximax tolerance τ on the scale of the regret-estimation error. The robustness studies of §7 show the recommendation is stable to θ and that quality is stable to nadir error.

When LUR is inappropriate. LUR targets the selection of a single robust alternative. It is not the tool when the DM genuinely wants to explore the whole trade-off surface, when criteria

are fundamentally incommensurable, or when reliable normalisation bounds cannot be obtained (the nadir is well known to be hard to estimate; our sensitivity study quantifies the resulting degradation).

Threats to validity. The held-out families, though disjoint from the probes, are themselves a modelling choice; we mitigate this by spanning linear, Chebyshev, ASF and CES forms and by reporting per-family results. Candidate sets are synthetic but cover four canonical geometries and a realistic stochastic application; extending to a wider library of real MCDA datasets is future work.

Future work. Natural next steps are ϵ -net guarantees showing that a finite probe family approximates a continuous admissible-utility class to $O(\epsilon)$; non-linear (product/ratio) probes for synergistic criteria; surrogate models for the direct continuous formulation in high dimension; and interactive elicitation that adapts the probe family from DM feedback, connecting LUR to interactive multiobjective optimisation.

9 Conclusion

We have presented the Leximax Universal-Regret core, a Pareto-compatible decision rule that replaces the efficient set as the final object of multicriteria analysis with an auditable regret certificate. LUR ranks alternatives by the lexicographically sorted vector of normalised regrets over a declared family of monotone probes, and returns a single robust recommendation together with an explanation of where it remains fragile. We proved existence and completeness, Pareto compatibility *under a separation condition*, dominated-solution exclusion, stability of the tolerance-based rule under bounded perturbations, and a disappointment-perturbation bound with explicit constants, and we gave a redundancy-aware, correlation-clustering probe construction that keeps the family linear in the number of criteria. A replicated held-out study over six preference families and a stochastic smart-grid case showed that LUR attains best-group worst-case robustness—practically equivalent to the strongest robust scalarisations and far ahead of the averaging and Monte-Carlo preference methods used in practice—while reducing sensitivity to redundant criteria, certifying its choice, and extending naturally to noisy criteria and multiple stakeholders.

The central message is that *Pareto efficiency is a safety property, not a complete decision rule*. By folding robustness across a declared class of questions into the selection itself, LUR turns an efficient set into a defensible, explainable recommendation—a low-regret compromise across the declared interpretations of the criteria, accompanied by a certificate that reports which questions remain most disappointed.

Reproducibility. All code, generated data, and scripts that regenerate every table and figure in this paper are released, including the four-geometry benchmark, the redundancy benchmark, the statistical tests, the sensitivity analyses, and the smart-grid case study.

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A Proofs

Theorem 1 (Existence and completeness). For finite A and finite $\mathcal{Q}$, each $D^\downarrow(x) \in [0, 1]^K$ is a well-defined real vector. The lexicographic order on $\mathbb{R}^K$ is reflexive, transitive and total, hence a complete preorder; since A is finite the lexicographic minimum exists. $\square$

Theorem 2 (Weak and strict Pareto compatibility). Let $x \prec_P y$, so $r_i(x) \leq r_i(y)$ for all i with strict inequality for some j . For any monotone non-decreasing probe q , $q(r(x)) \leq q(r(y))$; as q^*, q^w are constants independent of the alternative, $D_q(x) \leq D_q(y)$ for all q , giving $D^\downarrow(x) \leq D^\downarrow(y)$ componentwise after sorting and hence $x \preceq_{\text{LUR}} y$ (weak compatibility, monotonicity only). If $\mathcal{Q}$ separates criteria there is a probe q strictly increasing in r_j , for which $q(r(x)) < q(r(y))$ and thus $D_q(x) < D_q(y)$; the multiset of disappointments of x is then componentwise $\leq$ that of y with at least one strict drop, so the descending-sorted vectors satisfy $D^\downarrow(x) \leq_{\text{lex}} D^\downarrow(y)$ strictly and $x \prec_{\text{LUR}} y$. The counterexample in the main text shows the separation hypothesis cannot be dropped. $\square$

Corollary 1 (Dominated exclusion). If x is dominated by a feasible y and $\mathcal{Q}$ separates criteria, Theorem 2 gives $y \prec_{\text{LUR}} x$, so x is not a lexicographic minimiser. Without separation only $y \preceq_{\text{LUR}} x$ holds and a dominated alternative may tie its dominator. $\square$

Theorem 3 (Stability of the tolerance-based rule). Write $\hat{D}_q(x) = D_q(x) + \xi_q(x)$, $|\xi_q(x)| \leq \eta$; sorting is 1-Lipschitz in $\|\cdot\|_\infty$, so each sorted coordinate moves by at most η . Fix a rival y and the deciding coordinate $k = k(y)$ with true margin $D_{[k]}(y) - D_{[k]}(x^{\text{LUR}}) \geq \Delta$. Under perturbation this margin is at least $\Delta - 2\eta$; since $2\eta < \Delta - \tau$ it stays above τ , so the τ -tolerance rule still decides coordinate k in favour of x^{LUR} . For the earlier coordinates $j < k$, which agree within $\tau - 2\eta$ by hypothesis, the perturbed gaps stay within τ and are treated as ties, so the decision is not pre-empted before k . As $2\eta < \tau$ no spurious strict difference is created earlier. Hence x^{LUR} still beats every rival and the winner is unchanged. (Exact comparison, $\tau = 0$, is excluded precisely because the second condition $2\eta < \tau$ then fails.) $\square$

Theorem 4 (Non-negative-probe Pareto compatibility). If $q(r) = \sum_i w_i r_i$ with $w_i \geq 0$, then $r \leq r'$ componentwise implies $q(r) \leq q(r')$; likewise $q_{\max}(r) = \max_i w_i r_i \leq \max_i w_i r'_i = q_{\max}(r')$. All such probes are monotone non-decreasing, so Theorem 2 applies. $\square$

Lemma 1 (Disappointment perturbation). Fix q and its approximant q' with $\sup_r |q(r) - q'(r)| \leq \eta$; then also $|q^\star - q'^\star| \leq \eta$ and $|q^\text{w} - q'^\text{w}| \leq \eta$. Write the ranges $N = q^\text{w} - q^\star$ and $N' = q'^\text{w} - q'^\star$, both $\geq \beta > 0$. Using $D_q(x) \in [0, 1]$,

$$|D_q(x) - D_{q'}(x)| = \left| \frac{q(x) - q^\star}{N} - \frac{q'(x) - q'^\star}{N'} \right| \leq \frac{|q(x) - q'(x)| + |q^\star - q'^\star|}{N'} + D_q(x) \frac{|N - N'|}{N'} \leq \frac{2\eta + \eta}{\beta} = \frac{3\eta}{\beta}.$$

The constant is explicit and the dependence on the range floor β (worst case ε) is exhibited, making clear when the bound is informative. $\square$

Proposition 1 (Plug-in consistency). Let $\hat{\mu}_i$ be the mean of n i.i.d. joint draws, so $|\hat{\mu}_i - \mu_i| = O_p(n^{-1/2})$ by the law of large numbers and finite second moments. For L -Lipschitz probes $|\hat{q}(x) - q(x)| \leq L \sum_i |\hat{\mu}_i - \mu_i| = O_p(n^{-1/2})$, and since the normalisation anchors are continuous functionals of the estimated means, $|\hat{D}_q(x) - D_q(x)| = O_p(n^{-1/2})$. With finitely many candidates and probes a union bound gives a uniform error $\eta_n = O_p(n^{-1/2})$. Once $2\eta_n < \min\{\Delta - \tau, \tau\}$ the tolerance-rule stability (Theorem 3) preserves the winner; as $\eta_n \rightarrow 0$ this holds with probability tending to one. $\square$